

Attention-Aware Study Assistant

GenAI System Building Challenge

Microservices Architecture Presentation

System Overview

What It Is

A cloud-native, GenAI-powered NLP educational system built with **10 microservices**.

Core Function

Monitors a learner's **cognitive state** in real-time and dynamically adapts educational content delivery based on fatigue signals.

Modes of Operation

Learning Mode:

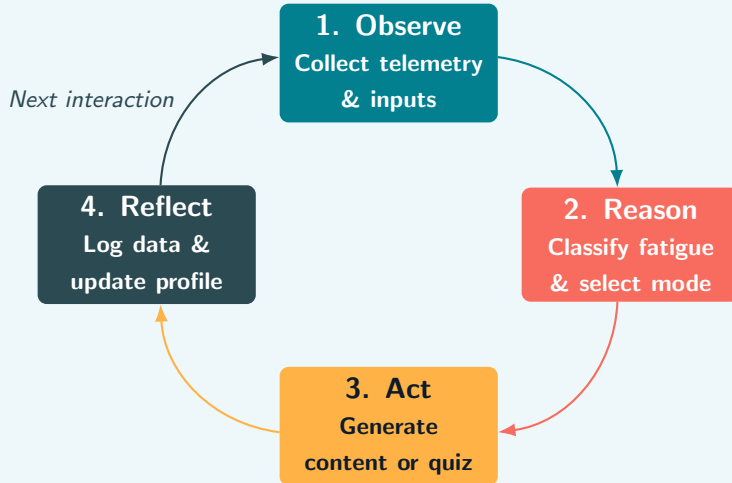
Video-based; fatigue from seek/pause/speed.

Quiz Mode: Text Q&A; fatigue from keypress & latency.

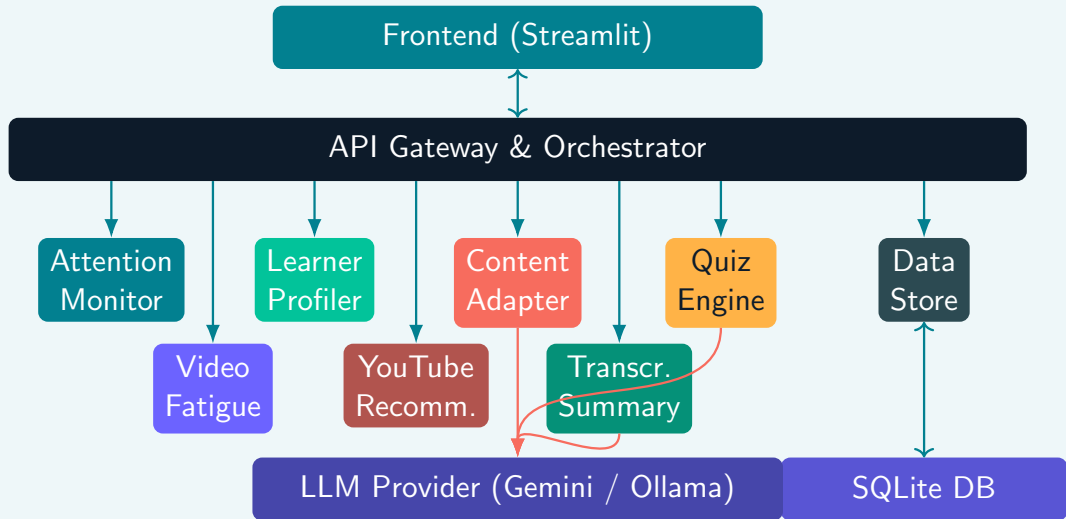
Learning Mode · Video-based · Fatigue inferred from seek, pause, speed changes **Quiz Mode** · Te

ReAct Agentic Loop

The system implements a **ReAct (Reason + Act)** loop where each interaction triggers a 4-step cognitive process:



Microservices Architecture

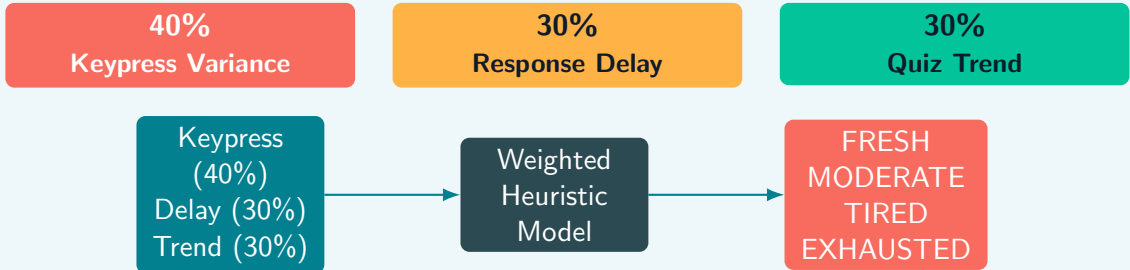


Attention Monitor Agent

Purpose: Passive fatigue classifier tracking cognitive load in real-time.

Fatigue Score Formula:

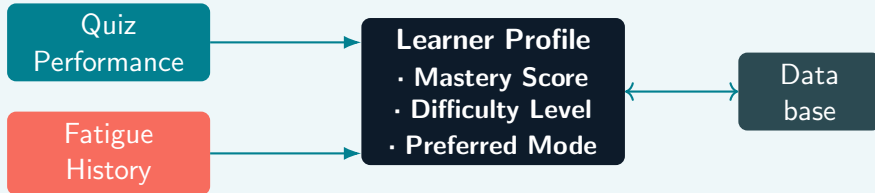
$$S_{\text{fatigue}} = 0.4 \left(\frac{\sigma_{\text{keypress}}}{\mu_{\text{baseline}}} \right) + 0.3 \left(\min \left(1, \frac{t_{\text{delay}}}{t_{\text{max}}} \right) \right) + 0.3 (1 - S_{\text{quiz_trend}})$$



Learner Profiler Agent

Purpose: Adaptive learner model maintenance over time.

- **Tracking:** Updates mastery score, difficulty level, and preferred mode based on continuous quiz performance and interaction history.



Content Adaptation Agent

Purpose: Dynamically selects teaching modes based on fatigue.



FRESH
Full Explanation

MODERATE
Guided Summary

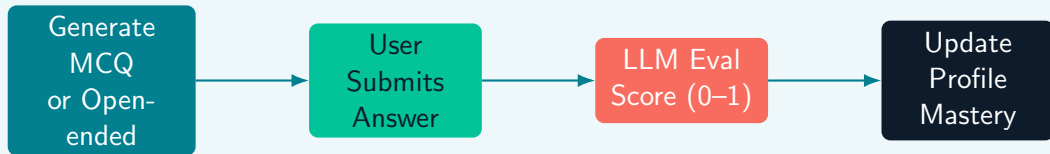
TIRED
Analogy Mode

EXHAUSTED
Break + Recap

Quiz Engine

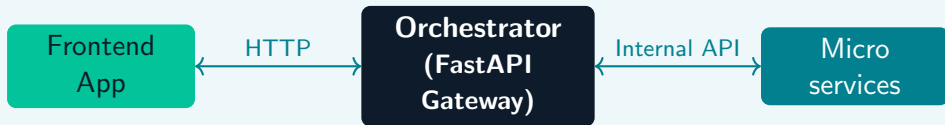
Purpose: Adaptive quiz generator & evaluator.

- **LLM Abstraction:** Auto-routes to Cloud API (Gemini) or Local Fallback (Ollama).
- **LLM-as-a-judge Scoring:** $S_{\text{score}} = \text{LLM_Evaluate}(\text{Answer}) \in [0.0, 1.0]$ — evaluates free-text on semantic correctness, updating the learner's mastery profile.



Orchestrator

Purpose: Central ReAct agentic loop controller routing all traffic.



FastAPI

High-performance async Python powering the gateway, enabling non-blocking inter-service communication.

HTTP Routing

Frontend communicates over HTTP REST; gateway dispatches to the relevant microservice via internal API.

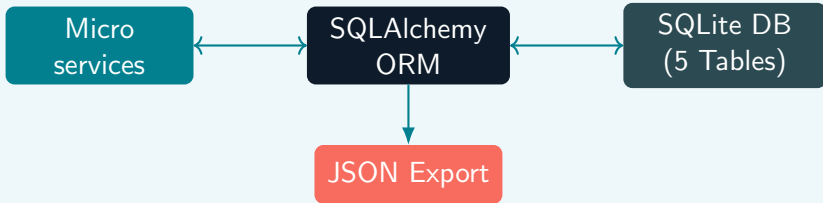
ReAct Control

Every incoming request triggers the full Observe→Reason→Act→Reflect cycle, coordinated here.

Data Persistence Service

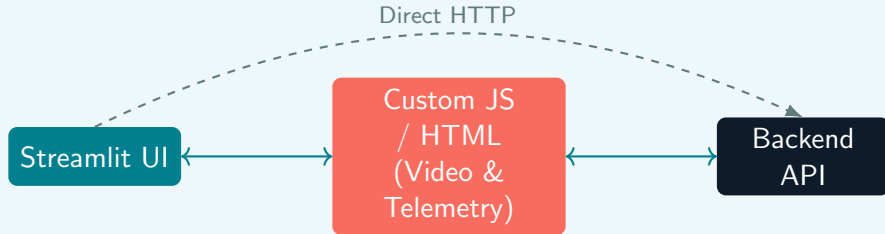
Purpose: Centralised storage and session export layer.

- **ORM (Object-Relational Mapping):** Abstracts raw SQL by allowing interaction with the database using Python objects/classes directly. Prevents SQL injection and provides a structured schema for our **5 tables**.



Frontend Application

Purpose: Streamlit interactive user interface with dual modes.



Learning Mode

Video player with real-time fatigue overlay; custom JS captures seek, pause, and speed events.

Quiz Mode

Interactive Q&A with keypress timing; monitors inter-key intervals and submission latency.

Session Dashboard

Live fatigue gauge, mastery progress bar, session history, and one-click JSON export.

Video Fatigue Monitor

Purpose: Maps player events to cognitive fatigue scores.

Score Calculation & Decay Mechanic:

$$F(t) = \sum_i \text{Impact}(E_i) \cdot e^{-\lambda(t-t_i)}, \quad \lambda = 0.02 \text{ (natural cognitive recovery rate)}$$

Event Impacts

■ Pause	+0.08
■ Seek Back	+0.06
■ Speed Down	+0.10
■ Speed Up	+0.04
■ Normal Play	—decay

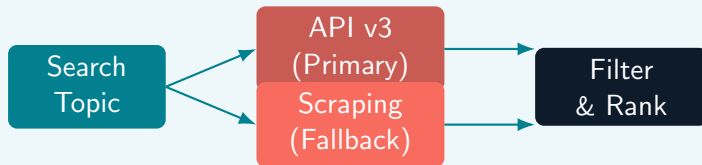
Blending Strategy

80% Local UI Tracking
Smooth, low-latency fatigue
gauge updates in browser

20% Server-side Compute
Accurate decay model and persistent event log

YouTube Recommender

Purpose: Fetches and ranks topic-relevant educational videos.

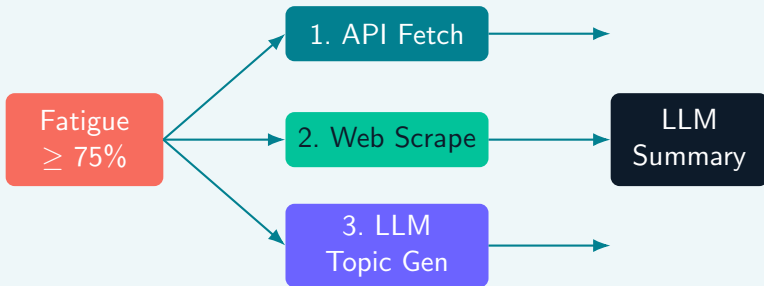


- Relevance scoring based on title & description keywords
- Duration filtering: prefers 5–20 min educational clips
- View count & rating used as quality proxies

Transcription Summary Agent

Purpose: Generates learner-friendly video summaries.

- **Trigger Threshold (0.75):** Intervenes precisely before cognitive exhaustion ($\geq 75\%$).
- **Pomodoro Dynamics:** Models the ~ 25 min human focus limit. Triggers a summary break just as attention starts to sharply decline.



Future Work

Adaptable Video Generation

Dynamically generating or editing video content to match learner states (e.g. leveraging SaralAI research on explaining papers via video).

Multi-modal Biometrics

Integrating webcam-based eye-tracking and facial expression analysis for richer, passive fatigue detection.

Advanced Cognitive Models

Scaling to more accurate machine learning algorithms for determining cognitive states beyond heuristics.

Curriculum Integration

Expanding beyond isolated topics to map knowledge gaps across a full semester's curriculum.

Cloud Deployment Strategy

Streamlit Cloud (Frontend)

Native, zero-config hosting for Streamlit; auto-deploys via GitHub.

Persistent Disk (SQLite)

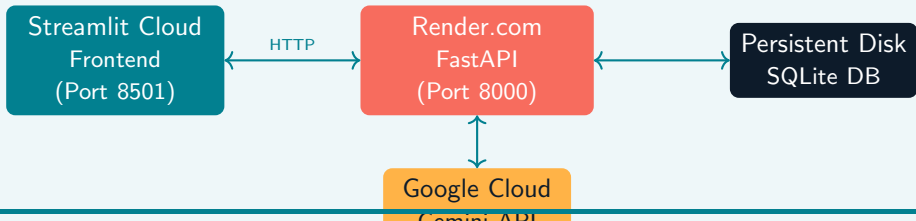
Render-managed volume providing durable storage across deployments.

Render.com (Backend API)

PaaS natively supporting FastAPI, managed persistent disks for SQLite, effortless horizontal scaling for 10 microservices.

Google Cloud (Gemini API)

Primary LLM provider; Ollama serves as local fallback for offline/privacy sessions.



What Worked Well

- Coding and implementation; rapid prototyping
- Initial ideation, requirements and use-case profiling
- Research regarding methods to use

Where It Fell Short

- First and foremost, **credits**
- Understanding complex ideas originally conceived by humans

Overall Impact

- Credit shortage was frustrating; highlighted dependency risk
- Hard realisation of development speed without GenAI
- What would take weeks was done in moments



Thank You

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Any questions?